

SerialDetector: Principled and Practical Exploration of Object Injection Vulnerabilities for the Web

SerialDetector: Principled and Practical Exploration of Object Injection Vulnerabilities for the Web - Author not stated, NDSS Symposium.

- Published: date not stated
- Original: <<https://www.ndss-symposium.org/ndss-paper/serialdetector-principled-and-practical-exploration-of-object-injection-vulnerabilities-for-the-web/>>
- Preserved from: <https://www.ndss-symposium.org/ndss-paper/serialdetector-principled-and-practical-exploration-of-object-injection-vulnerabilities-for-the-web/> (live) on 2026-08-08
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The last decade has seen a proliferation of code-reuse attacks in the context of web applications. These attacks stem from Object Injection Vulnerabilities (OIV) enabling attacker-controlled data to abuse legitimate code fragments within a web application's codebase to execute a code chain (gadget) that performs malicious computations, like remote code execution, on attacker's behalf. OIVs occur when untrusted data is used to instantiate an object of attacker-controlled type with attacker-chosen properties, thus triggering the execution of code available but not necessarily used by the application. In the web application domain, OIVs may arise during the process of deserialization of client-side data, e.g., HTTP requests, when reconstructing the object graph that is subsequently processed by the backend applications on the server side.

This paper presents the first systematic approach for detecting and exploiting OIVs in .NET applications including the framework and libraries. Our key insight is: The root cause of OIVs is the untrusted information flow from an application's public entry points (e.g., HTTP request handlers) to sensitive methods that create objects of arbitrary types (e.g., reflection APIs) to invoke methods (e.g., native/virtual methods) that trigger the execution of a gadget. Drawing on this insight, we develop and implement SerialDetector, a taint-based dataflow analysis that discovers OIV patterns in .NET assemblies automatically. We then use these patterns to match publicly available gadgets and to automatically validate the feasibility of OIV attacks. We demonstrate the effectiveness of our approach by an in-depth evaluation of a complex production software such as the Azure

DevOps Server. We describe the key threat models and report on several remote code execution vulnerabilities found by SerialDetector, including three CVEs on Azure DevOps Server. We also perform an in-breadth security analysis of recent publicly available CVEs. Our results show that SerialDetector can detect OIVs effectively and efficiently. We release our tool publicly to support open science and encourage researchers and practitioners explore the topic further.

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